

Geometry, Topology, and Physics – Mikio Nakahara

Notes by Gene

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1 Quantum Physics

1.1 Analytical Mechanics

1.1.1 Newtonian Mechanics

Equation of motion: $m \frac{d^2 x}{dt^2} = F(x)$

If $F(x) = -\nabla V(x)$, then $V(x)$ is called the **potential** and F is called a **conserved force**. The quantity

$$E = \frac{m}{2} \left(\frac{dx}{dt} \right)^2 + V(x)$$

is conserved: $dE/dt = 0$.

1.1.2 Lagrangian Mechanics

Describe the system with N parameters $\{q_i\}$, the **generalized coordinates**. The generalized velocity is $\dot{q} = dq/dt$.

Let there be a Lagrangian function $L(q, \dot{q})$. Define the action as a functional

$$S[q(t), \dot{q}(t)] = \int_{t_i}^{t_f} L(q, \dot{q}) dt$$

Hamilton's principle: the physical trajectory is that of the least action. Consider a trajectory $q(t)$ of a

particle and vary it while holding endpoints fixed; $\delta q(t_i) = \delta q(t_f) = 0$.

$$\begin{aligned}
 0 = \delta S &= \int_{t_i}^{t_f} L(q + \delta q, \dot{q} + \delta \dot{q}) dt - \int_{t_i}^{t_f} L(q, \dot{q}) dt \\
 &= \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \frac{d(\delta q)}{dt} \right) dt \\
 &= \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) \delta q dt + \underbrace{\left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_i}^{t_f}}_0
 \end{aligned}$$

Thus we get the Euler-Lagrange equation

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0$$

Define the **generalized momentum** as $p = \partial L / \partial \dot{q}$. Then Euler-Lagrange becomes $\dot{p} = \partial L / \partial q$.

If L is independent of q , then q is called **cyclic**. The momentum conjugate to a cyclic coordinate is conserved:

$$\frac{\partial L}{\partial q} = 0 \longrightarrow \frac{dp}{dt} = 0$$

1.1.3 Hamiltonian Mechanics